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CoWiC-MAC: A coordinated WiFi-centric MAC mechanism in heterogeneous wireless networks☆

Tao Cheng [a], Shining Li [a,*], Yan Pan [b]

[a] School of Computer Science, Northwestern Polytechnical University, Xi'an, 710129, China
[b] Science and Technology on Information Systems Engineering Laboratory, National University of Defense Technology, Changsha, 410003, China

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ABSTRACT

The surging density of heterogeneous wireless networks in the 2.4 GHz ISM band has intensified cross-technology interference, particularly between high-power WiFi and low-power ZigBee. Current coexistence strategies primarily rely on opportunistic access or rigid "all-or-nothing" channel reservation. However, they inherently fail to balance the disparate communication demands of both protocols, often resulting in severe transmission unfairness and uncoordinated packet collisions under heavy traffic loads. In this paper, we propose *CoWiC-MAC*, a systematic coordination framework that shifts coexistence from passive contention to proactive, demand-driven coordination. Leveraging the enhanced computational capability of WiFi Access Points (APs) and the bidirectional feedback enabled by Cross-Technology Communication (CTC), CoWiC-MAC transforms the AP into an intelligent orchestrator. Specifically, a Mixed-Integer Linear Programming (MILP)-based scheduling model is developed to reorder WiFi traffic, creating interference-free transmission opportunities for ZigBee while bounding the induced scheduling overhead. In parallel, a lightweight Link-Weighted Superframe Adaptation (L-WSA) algorithm is designed at the ZigBee coordinator to adapt transmission parameters based on fine-grained channel feedback. Extensive trace-driven evaluations demonstrate that CoWiC-MAC achieves a ZigBee transmission gain of up to 96.3% while maintaining zero WiFi frame group loss and controlled delay overhead. Compared to state-of-the-art approaches, which achieve only up to 33.3% ZigBee gain while still incurring non-negligible WiFi loss, CoWiC-MAC provides a more balanced and robust coexistence solution. These results highlight its effectiveness in enabling efficient and reliable cross-technology coordination in dense and heterogeneous IoT environments.

1. Introduction

The proliferation of wireless communication technologies has led to an unprecedented demand for the unlicensed 2.4 GHz ISM band [1–3]. In this shared spectrum, diverse technologies — most notably WiFi and ZigBee — must coexist and compete for limited spectrum resources. This coexistence often triggers severe Cross-Technology Interference (CTI) [4], resulting in a "lose-lose" situation: empirical studies [5,6] show that while ZigBee's packet loss rate can skyrocket to over 60%, WiFi simultaneously suffers a 20% throughput degradation due to frequent backoff and retransmissions. Such mutual interference significantly degrades communication efficiency, potentially jeopardizing mission-critical IoT applications.

The root of these CTI challenges stems from two fundamental asymmetries. First, the temporal stochasticity of WiFi traffic — being inherently bursty — leaves fragmented and unpredictable whitespaces for other technologies [7,8]. Second, by design, WiFi employs higher transmission power and more aggressive channel access mechanisms [9]. Without inter-technology coordination, this dominance allows WiFi to effectively suppress the communication opportunities of coexisting ZigBee nodes, leading to severe resource starvation.

Facing these persistent challenges, the rapid evolution of WiFi standards — notably IEEE 802.11ax [10,11] — has primarily focused on improving intra-network efficiency rather than mitigating coexistence issues. While advanced features such as Trigger-based Target Wake Time (TWT) and OFDMA enhance WiFi performance [12,13], they remain agnostic to heterogeneous neighbors, leaving CTI largely unaddressed. Meanwhile, this evolution has significantly enhanced the computational capability and scheduling granularity of modern WiFi APs [14], providing a critical foundation for cross-protocol coordination. APs now possess enhanced processing intelligence to host complex

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* Corresponding author.
E-mail addresses: nwpu_taocheng@hotmail.com (T. Cheng), lishining@nwpu.edu.cn (S. Li), panyan@nudt.edu.cn (Y. Pan).

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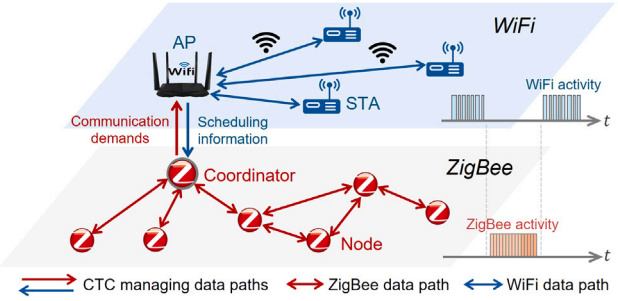


Fig. 1. Time-division channel access via demand interaction between WiFi AP and ZigBee coordinator.

cross-protocol scheduling tasks [15], offering a previously unavailable platform for centralized resource management.

To bridge the communication gap between heterogeneous protocols, Cross-Technology Communication (CTC) [16–21] has recently emerged as a disruptive paradigm. By enabling direct information exchange between heterogeneous devices — such as WiFi and ZigBee — CTC provides a signaling channel for precise resource negotiation without requiring additional hardware gateways. Building upon this capability and the enhanced AP intelligence, we argue that WiFi APs are well-positioned to serve as centralized coordinators within the shared spectrum. As illustrated in Fig. 1, we envision a WiFi-centric, bidirectional coordination framework in which the AP interacts with the ZigBee Coordinator (ZC) via CTC to exchange uplink communication demands and downlink scheduling information. Based on this interaction, the AP can adapt its transmission behavior to create transmission opportunities for ZigBee, enabling a form of time-division channel access between heterogeneous networks. This explicit bidirectional coordination provides a foundation for more efficient and balanced coexistence.

However, translating CTC-based coordination into a robust and efficient MAC protocol entails two fundamental challenges. First is the lack of a QoS-aware and system-level coordination framework for cross-technology orchestration. Most current CTC-based studies focus on isolated link-layer interactions, treating heterogeneous networks as "black boxes" without a system-level coordination perspective. Specifically, pioneering explicit coordination techniques are limited by their localized paradigms: ECC [22] relies on "blind" channel reservation without awareness of ZigBee's actual demand, while BiCord [23] relies on multi-round negotiation to convey demands and allocates resources without explicitly considering the impact on WiFi traffic. Achieving seamless coexistence requires moving beyond such localized interactions toward a global viewpoint that can harmonize the disparate characteristics of both protocols. Second is the dilemma of balancing mutual performance requirements—the "zero-sum" trap. Prior studies [24,25] often adopt a "master–slave" logic where WiFi must completely vacate the channel. Such approaches fail to consider the intrinsic trade-offs between WiFi's high-throughput demands and ZigBee's reliability needs. Developing a mechanism that minimizes WiFi's traffic disruption while ensuring ZigBee's adaptive survival within restricted, non-guaranteed windows remains a challenging task.

To bridge these gaps, this paper proposes CoWiC-MAC, a systematic, WiFi-centric coordinated MAC mechanism designed for efficient coexistence. Unlike traditional "unilateral suppression" approaches, CoWiC-MAC transforms the WiFi AP into an intelligent orchestrator that proactively manages the shared spectrum through a two-stage synergistic optimization strategy. Specifically, the AP first establishes a coordination link with the ZigBee network to acquire its real-time communication demands. Based on these demands and the current WiFi traffic load, the AP adaptively defers a portion of WiFi transmissions to reserve dedicated transmission windows for ZigBee, subsequently feeding this allocation information back to the ZC. Upon receiving the feedback, the ZC adaptively reshapes its superframe structure to maximize transmission gains within the allocated resources. The mechanism comprises three key components: (1) Connection Establishment: The AP utilizes cross-technology beacons to synchronize with ZigBee and opens specific sensing windows to capture transmission demands directly from the ZC via CTC methods. (2) Proactive Traffic Orchestration: The AP employs an optimization algorithm to reserve temporal transmission windows for ZigBee by minimizing the weighted sum of WiFi Frame Group (FG) loss and latency, ensuring that WiFi's own Quality of Service (QoS) remains within acceptable bounds. (3) Localized Superframe Adaptation: To cope with potentially limited resources, the ZC dynamically reconfigures its CAP and CFP slot durations, aligning its active communication periods with the specific windows allocated by the AP to maximize transmission gain.

In summary, this paper makes the following contributions:

- We design CoWiC-MAC, a systematic framework that achieves fine-grained resource orchestration between WiFi and ZigBee via direct CTC.
- We develop a joint-optimization-inspired scheduling logic that balances the QoS requirements of both high-power and low-power technologies, moving beyond the simple "all-or-nothing" channel access.
- We validate CoWiC-MAC via NS-3 simulations with real-world WiFi traces. Results show a ZigBee transmission gain of 96.3% (vs. 33.3% in SOTA) and a zero WiFi frame group loss rate (vs. 0.9% in SOTA and 6.8% in baseline), effectively resolving the coexistence dilemma even under near-saturation traffic loads.

The remainder of this paper is structured as follows: Section 2 reviews related work; Section 3 outlines the characteristics of wireless coexistence technologies and the principles of CTC methods; Section 4 details the CoWiC-MAC design; Section 5 presents the experimental evaluation and performance analysis; and Section 6 concludes the paper.

2. Related work

In environments where heterogeneous wireless networks coexist, CTI is inevitable due to the uncoordinated access to the shared medium. To address this, prior research has primarily focused on enhancing error resilience or partitioning communication resources in the time or frequency domain. These solutions are generally categorized into two classes: interference mitigation and interference avoidance.

Interference mitigation mechanisms aim to improve packet decoding accuracy despite the presence of interference [26–29]. Techniques such as forward error correction, adaptive retransmissions, and advanced signal processing are commonly employed. For example, BuzzBuzz [27] mitigates WiFi interference in both asymmetric and symmetric regions by applying forward error correction and a multi-header technique based on statistical analysis of corrupted packet fields. CrossZig [28] reconstructs corrupted ZigBee packets by stitching together correct symbol fragments or merging retransmitted signals, leveraging physical-layer characteristics. Similarly, PolarScout [29] exploits the signal disparity between protocols by applying a moving average over the standard half-sine waveforms, effectively extracting the distinctive signatures of ZigBee signals from collisions. While effective, these approaches often introduce high computational overhead and latency, making them less suitable for power-constrained IoT devices or real-time applications.

Interference avoidance mechanisms prevent CTI by sensing interfering signals and proactively adjusting communication timing or frequency [7,30–34]. These approaches can be classified as non-cooperative or cooperative, depending on whether cross-technology coordination is involved. In non-cooperative schemes, devices independently detect interference and adapt their behavior without interacting across technologies. For instance, WISE [7] models WiFi traffic whitespace by

Table 1
Comparison of cross-technology coordination approaches.

Approach	Coordination paradigm	Demand acquisition	WiFi impact awareness	Opt. scope
ECC [22]	Unidirectional Notice	Implicit/None	Low (Heuristic)	Local
BiCord [23]	Bidirectional Aware	One-bit/Multi-round	Low (Reactive)	Local
CoWiC-MAC	**Bidirectional Synergy**	**Direct**	**High (Joint-opt.)**	Global

sampling RSSI on ZigBee devices and adapts frame lengths based on predicted channel availability. EmBee [35] detects ZigBee activity from the WiFi side by identifying shifts in signal center frequency, thereby enabling spectral separation through subcarrier reservation. However, the absence of explicit coordination often leads to suboptimal performance in highly dynamic environments. Cooperative approaches utilize cross-technology signaling to coordinate channel access. CBT [30] enables ZigBee nodes to proactively transmit busy signals during their transmissions, increasing visibility to WiFi and improving contention fairness. WiCop [31] introduces a gateway that transmits fake WiFi preambles to reserve the channel for ZigBee. Although these methods enhance visibility between technologies, they often rely on specialized hardware or incur significant overhead by mimicking foreign standard preambles.

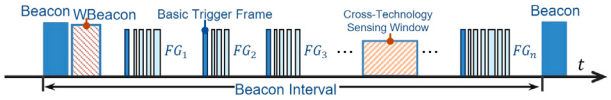
Recently, CTC [16–21] has emerged as a promising solution for enabling explicit collaboration among heterogeneous wireless networks [22,23,36,37]. By allowing direct exchange of communication demands between devices using different technologies, CTC provides a foundation for proactive and fine-grained interference management. For instance, ECC [22] utilizes the CTS mechanism supported by WiFi to reserve the channel and employs CTC to notify ZigBee devices, enabling coordinated access while maintaining WiFi performance. BiCord [23] focuses on expressing ZigBee's communication demands through CTC and introduces a bidirectional coordination mechanism that allows WiFi to allocate resources to ZigBee as needed. CTC-CSMA/CA [24] further integrates CTC with the CSMA/CA protocol, enabling ZigBee devices to actively contend for channel access alongside WiFi. These studies demonstrate the potential of CTC to facilitate harmonious coexistence through explicit demand exchange. However, a critical research gap remains: most existing CTC-based strategies prioritize IoT protection through a "best-effort" resource allocation logic, often implicitly assuming that WiFi can yield resources without explicitly accounting for its QoS impact. Furthermore, these schemes treat ZigBee as a "black boxes", overlooking MAC-layer characteristics (e.g., superframe structures), which leads to misaligned resource windows and suboptimal spectrum efficiency. Table 1 provides a comparative summary, highlighting the architectural advancement of CoWiC-MAC over prior explicit coordination frameworks.

Unlike previous works, CoWiC-MAC adopts a QoS-aware coordination strategy that jointly optimizes the performance of both WiFi and ZigBee networks, rather than favoring one side at the expense of the other. CoWiC-MAC builds upon CTC-based signaling to establish a WiFi-centric coordination framework, where the AP acts as an intelligent orchestrator to capture real-time ZigBee spectrum demands. By jointly considering cross-technology demand, WiFi performance constraints, and ZigBee superframe characteristics, CoWiC-MAC enables fine-grained resource allocation through coordinated WiFi traffic deferral and ZigBee superframe adaptation, achieving efficient and balanced coexistence.

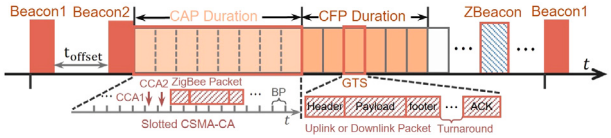
3. Preliminary

3.1. Coexistence technology characteristics

WiFi Activity: Unlike traditional CSMA/CA-based random access, IEEE 802.11ax and its successors [11] introduce deterministic scheduling features such as TWT and OFDMA. These technologies endow the WiFi AP with fine-grained temporal control over the transmission behavior of associated stations (STAs), establishing the protocol



(a) Basic WiFi Operation Timing Diagram in CoWiC-MAC



(b) Basic ZigBee Operation Timing Diagram in CoWiC-MAC

Fig. 2. Timing Diagram of Coexistence Technology Operations.

foundation for AP-centric traffic orchestration in CoWiC-MAC. APs supporting 802.11ax can support the integration of CTC within this framework while remaining transparent to STAs. Subsequent designs are also developed based on the 802.11ax standard. As illustrated in Fig. 2(a), within a typical beacon interval, the AP broadcasts a standard beacon followed by a WBeacon—a specialized CTC signal constructed to notify IoT nodes of coexistence parameters, such as network metadata and sensing window configurations. During the predefined cross-technology sensing window, the AP listens for access demands from IoT devices. For intra-network WiFi communication, the AP employs a trigger-based TWT mechanism: it first establishes a global time reference via beacon broadcasting and then utilizes Basic Trigger Frames to schedule uplink and downlink transmissions for multiple clients. Authorized clients are allowed to transmit frame groups (FGs) following the Basic Trigger Frames. This mechanism regulates WiFi traffic in the time domain, creating deterministic channel opportunities for heterogeneous IoT devices.

ZigBee Activity: For the ZigBee subnet operating in beacon-enabled mode, communication is centrally scheduled by the ZC [38]. In a standard superframe structure, the ZC periodically broadcasts a beacon to disseminate network management information, forming a complete active period composed of the Contention Access Period (CAP) and the Contention-Free Period (CFP). During this superframe, nodes access the channel via either slotted CSMA/CA in the CAP or Guaranteed Time Slots (GTS) in the CFP. As illustrated in Fig. 2(b), CoWiC-MAC introduces a flexible restructuring of the conventional beacon mechanism. Specifically, the beacon functionality is logically decoupled into Beacon1 and Beacon2. Beacon1 periodically indicates the time offset of Beacon2, enabling nodes to wake up accordingly, while Beacon2 carries detailed scheduling information such as time-slot allocation and management commands.

Furthermore, the ZC conveys real-time traffic demands to the WiFi AP by transmitting a ZBeacon—a specialized ZigBee-compliant data frame designed for CTC. This signal can be detected by the WiFi AP in spectrum sensing mode while remaining transparent to conventional ZigBee nodes. The transmission of ZBeacon is scheduled to fall within the sensing window reserved by the WiFi AP, enabling on-demand information exchange and coordinated resource orchestration across heterogeneous networks.

3.2. Cross-technology communication

Cross-technology communication from WiFi to ZigBee (W2Z) has become a widely adopted and well-studied approach. By selectively

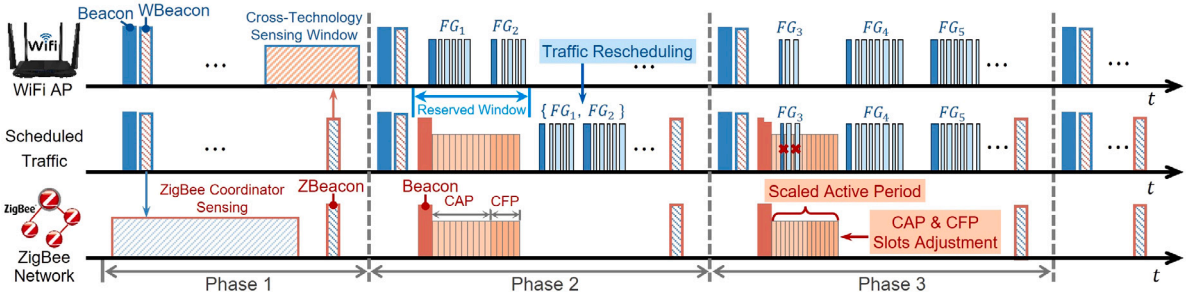


Fig. 3. CoWiC Basic Idea: Enabling Collaborative Channel Access via WiFi Traffic Rescheduling and Adaptive ZigBee Transmission.

modifying the payload of WiFi packets, signal emulation techniques [16] can generate waveforms in the time domain that closely resemble standard ZigBee signals, thereby enabling direct physical-layer communication. Furthermore, as ZigBee utilizes Direct Sequence Spread Spectrum (DSSS) modulation, partial errors within the emulated signals do not compromise the final decoding accuracy. Recent studies [19,20] demonstrate that CTC signals emulated by WiFi can achieve performance nearly identical to standard ZigBee signals in terms of both throughput and packet reception rate (PRR), and can even improve robustness under certain channel conditions due to WiFi's higher transmission power. In this work, CoWiC-MAC adopts the WiCast [20] framework to implement downlink coordination, ensuring reliable and compatible scheduling information delivery from the WiFi AP to the ZC.

Unlike the W2Z direction, emulating complex high-sampling-rate WiFi signals using low-sampling-rate ZigBee signals is inherently infeasible. Prior research [18,39] addresses this by constructing specific ZigBee payloads to generate identifiable signal signatures at the WiFi receiver, enabling indirect ZigBee-to-WiFi (Z2W) communication. Specifically, when a ZigBee device transmits specific symbol patterns (e.g., "0xEF" and "0x67"), its signal can induce stable phase shifts of $+4\pi/5$ and $-4\pi/5$ at the WiFi receiver, respectively. Although these symbols cannot be directly decoded as standard WiFi frames, the resulting phase patterns can be extracted through signal processing at the WiFi side. By encoding information into such phase signatures, Z2W CTC can be achieved at 1/8 of the standard ZigBee data rate, which is sufficient for conveying compact demand metadata. In this paper, we adopt the SymBee [39] methodology for CoWiC-MAC. The ZC constructs ZBeacons using these specialized symbols, while the WiFi AP performs signal sensing and decoding during the predefined cross-technology sensing windows.

4. Designs

In this section, we present the design of CoWiC-MAC, a WiFi-centric framework that enables coordinated channel access in heterogeneous wireless environments. We first provide an overview of the system architecture, and then introduce the key mechanisms that realize cross-technology coordination between WiFi and ZigBee.

4.1. Overview

To mitigate contention-induced performance degradation in heterogeneous wireless coexistence, we propose CoWiC-MAC, a WiFi-centric coordination framework that shifts channel access from stochastic contention to proactive coordination. CoWiC-MAC operates in a continuous closed-loop process between the WiFi AP and the ZigBee Coordinator (ZC). Through direct cross-technology information exchange, the two entities iteratively exchange demand and scheduling information to align their network operations. As illustrated in Fig. 3, the overall workflow consists of three phases:

Phase 1: Cross-Technology Connection Establishment. The WiFi AP initiates coordination by broadcasting a *WBeacon*, which conveys

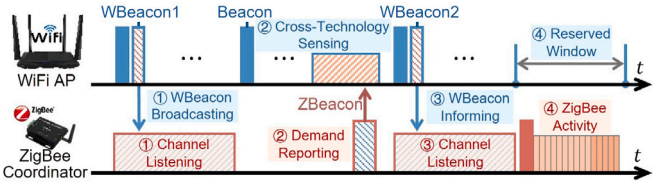


Fig. 4. Schematic of WiFi-ZigBee connection establishment.

network metadata and defines a cross-technology sensing window for the coexisting ZigBee network. The ZC passively monitors the channel during its idle periods and, upon receiving the WBeacon, acquires both the AP's coordination parameters and the timing of the sensing window. Within this designated window, the ZC transmits a *ZBeacon* to report the communication demands of the ZigBee subnet. This interaction establishes a bidirectional coordination link between the AP and the ZigBee network, enabling explicit demand reporting prior to scheduling decisions.

Phase 2: WiFi Traffic Rescheduling. Based on the acquired ZigBee demand and WiFi traffic conditions, the AP performs traffic rescheduling based on an optimization-driven mechanism, aiming to reserve transmission opportunities for ZigBee while maintaining acceptable WiFi performance. Specifically, a portion of WiFi frame groups is strategically deferred to create an interference-free window (denoted as the Reserved Window in Fig. 3) for ZigBee communication. The resulting scheduling decisions, including the allocation of this transmission window, are encapsulated in subsequent WBeacons and communicated back to the ZC.

Phase 3: ZigBee Adaptive Transmission. Due to the inherent trade-offs between WiFi and ZigBee performance, the transmission windows allocated by the AP may still contain residual WiFi traffic. To cope with such partial interference, we design an adaptive ZigBee transmission mechanism featuring scaled active periods and dynamic CAP/CFP slot adjustments. Based on the residual interference information conveyed by the AP, the ZC adjusts its superframe configuration to better align with the available interference-free budget, thereby improving transmission efficiency under constrained conditions.

Moreover, CoWiC-MAC also has the potential to assist the ZC in facilitating data distribution across the ZigBee network [40], further improving overall network efficiency. Together, these components form a closed-loop coordination framework. The detailed designs of these core components—Cross-Technology Connection Establishment, WiFi Traffic Rescheduling, and ZigBee Adaptive Transmission—are presented in Sections 4.2, 4.3, and 4.4, respectively.

4.2. Cross-technology connection establishment

Robust connection establishment is the cornerstone of CoWiC-MAC, enabling bidirectional exchange of communication demands and scheduling information. By forming a coordination link between the WiFi

AP and the ZC, the ZigBee subnet can report its communication requirements, while the WiFi AP can subsequently provide scheduling guidance. As illustrated in Fig. 4, the connection establishment process follows a structured four-step interaction:

Step 1: Network Discovery and Synchronization. The WiFi AP periodically broadcasts a cross-technology beacon (WBeacon 1) to disseminate network metadata, including the WBeacon interval and the designated cross-technology sensing windows. Since the ZC operates in a default channel-listening mode, it can detect the WBeacon without prior association. Upon reception, the ZC obtains a coarse timing reference from the AP and prepares a ZBeacon to report its communication requirements based on the current ZigBee superframe configuration.

Step 2: Demand Reporting and Link Initialization. Within the cross-technology sensing window specified in WBeacon 1, the WiFi AP switches to spectrum sensing mode to perform cross-technology signal detection. During this period, the ZC transmits a ZBeacon using the Z2W CTC primitive to report its communication demands, including traffic load and timing requirements. Upon successful decoding of the ZBeacon (or in subsequent coordination cycles if necessary), a coordination link between the WiFi AP and the ZigBee subnet is established. At this point, the AP has acquired the necessary information to support subsequent cross-technology coordination.

Step 3: Configuration Notification. Based on the reported ZigBee demands, the WiFi AP performs traffic rescheduling and broadcasts the next cross-technology beacon (WBeacon 2) according to the synchronized cycle. This beacon conveys the allocated interference-free window and related scheduling information to the ZC. As the ZC continuously monitors the channel, it can promptly receive the WBeacon 2 and adjust its internal superframe scheduling to align ZigBee transmissions with the notified window.

Step 4: Coordinated Collision-Free Access. Within the reserved interference-free window, the ZigBee nodes perform network activities according to the adapted superframe schedule, while the WiFi network suppresses or defers its transmissions to maintain channel availability. This coordinated operation enables both networks to achieve predictable and controlled access to the shared spectrum.

The above four-step interaction establishes a bidirectional coordination link, enabling the exchange of demand and scheduling information between the WiFi AP and the ZigBee subnet. Building upon this foundation, the subsequent design focuses on how these parameters are utilized for efficient coexistence.

4.3. WiFi traffic rescheduling

The WiFi traffic rescheduling module enables the AP to proactively reshape channel access in the time domain based on the communication demands of the ZigBee subnet. Instead of reducing WiFi traffic, the AP reschedules transmissions to transform fragmented channel usage into structured patterns.

Specifically, leveraging the fine-grained scheduling capability of IEEE 802.11ax, the AP operates on *frame groups (FGs)*—each representing an aggregated burst of WiFi packets from a single STA within a short interval (e.g., sub-millisecond timescale). This abstraction aligns with the trigger-based TWT mechanism and significantly reduces scheduling complexity compared to packet-level control. By aggregating and temporally aligning these FGs, the AP consolidates scattered idle intervals into a contiguous interference-free window for ZigBee.

As illustrated in Fig. 5, once the ZigBee communication demand is acquired via ZBeacon reporting, the AP performs traffic rescheduling for the upcoming scheduling cycle. The objective is to carve out a flexible interference-free window for ZigBee data transmission, while reserving a fixed sensing window for reliable cross-technology demand reporting. Concretely, the AP selectively defers and reorders WiFi FGs to eliminate their overlap with the reserved windows. The resulting interference-free interval is then announced to the ZC through the subsequent WBeacon, enabling aligned and collision-free channel access

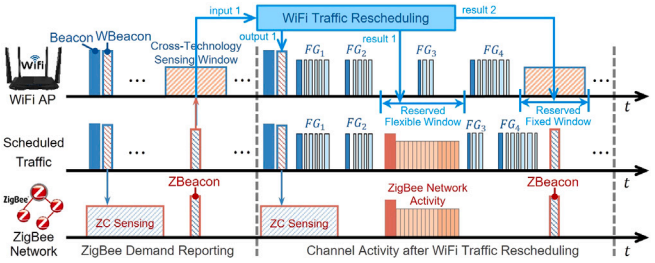


Fig. 5. Schematic of WiFi traffic rescheduling.

Table 2
Key variables in WiFi traffic rescheduling.

Variables	Definitions
T_{start}, T_{end}	Start and end of the scheduling horizon
$T_{\text{free}}^{start}, T_{\text{free}}^{end}$	Interference-free window for ZigBee activity
$T_{\text{sensing}}^{start}, T_{\text{sensing}}^{end}$	Sensing window for cross-technology demand reporting
D_{free}	Duration of interference-free window
D_{sensing}	Duration of cross-technology sensing window
K	Number of WiFi FGs to be scheduled
a_k	Arrival time of FG k
d_k	Duration of FG k
e_k	Deadline of FG k
s_k	Original transmission time of FG k
s'_k	Scheduled transmission time of FG k
y_k	Scheduling delay of FG k ($y_k = s'_k - s_k$)
P_k	Binary: 1 if FG k is postponed to the next cycle
μ_k	Dynamic postponement cost
γ	Penalty weight for sensing window protection
λ	Penalty weight for scheduling deviation

between the two networks. This coordination is performed at each scheduling cycle, allowing dynamic adaptation to traffic variations.

This design introduces a fundamental trade-off: enlarging the reserved window requires deferring more WiFi FGs, which may degrade latency performance, whereas strictly preserving WiFi timing limits the available airtime for ZigBee. Therefore, the AP must jointly determine the scheduling of WiFi FGs and the placement of ZigBee-related windows to balance cross-technology coordination and WiFi QoS.

To formalize the above scheduling process, we model WiFi traffic rescheduling as a constrained optimization problem over a scheduling horizon $[T_{start}, T_{end}]$. The parameters and decision variables are summarized in Table 2. Within this interval, the AP schedules a set of K WiFi frame groups (FGs), each characterized by its arrival time a_k, transmission duration d_k, and deadline e_k. The objective is to jointly determine (i) the scheduled transmission time s'_k for each FG, and (ii) the placement of ZigBee-related windows, such that cross-technology interference is minimized while WiFi QoS degradation remains controlled.

The optimization objective is formulated as:

$$\begin{aligned}\min \sum_{k=1}^{K} \mathbb{1}\left([s'_k,\ s'_k + d_k] \cap [T_{\text{free}}^{start},\ T_{\text{free}}^{end}] \neq \emptyset\right) \cdot d_k \\ + \gamma \sum_{k=1}^{K} \mathbb{1}\left([s'_k,\ s'_k + d_k] \cap [T_{\text{sensing}}^{start},\ T_{\text{sensing}}^{end}] \neq \emptyset\right) \cdot d_k \\ + \lambda \sum_{k=1}^{K} |s'_k - s_k| + \sum_{k=1}^{K} \mu_k \cdot \mathbb{1}(P_k = 1).\end{aligned} \tag{1}$$

The objective function reflects four key aspects of the scheduling trade-off: **(1) Interference to ZigBee transmission window:** The first term minimizes the total duration of WiFi FGs overlapping with the interference-free window. **(2) Protection of sensing window:** The second term penalizes overlaps with the sensing window using a higher weight γ, ensuring reliable cross-technology demand reporting. **(3)**

Temporal deviation of WiFi FGs: The third term penalizes deviations from the original transmission schedule, thereby controlling delay and jitter introduced by rescheduling. **(4) Deadline-aware postponement:** The final term penalizes deferring FGs to the next scheduling cycle, with a dynamically adjusted cost based on urgency, defined as:

$$\mu_k = \mu_{\text{base}} \cdot \frac{1}{\max\left(\epsilon, e_k - (T_{\text{end}} + d_k)\right)}. \tag{2}$$

This adaptive cost structure prioritizes delay-sensitive traffic by assigning higher postponement penalties to FGs approaching their deadlines, while allowing more flexibility for delay-tolerant traffic.

Beyond the objective function, the feasibility of the scheduling solution is governed by three categories of constraints.

(1) Window Timing Constraints: The ZigBee sensing window $[T^{start}_{\text{sensing}}, T^{end}_{\text{sensing}}]$ and interference-free window $[T^{start}_{\text{free}}, T^{end}_{\text{free}}]$ must lie within the scheduling horizon $[T_{start}, T_{end}]$ and remain non-overlapping. Their positions are defined by:

$$\begin{aligned} T^{end}_{\text{free}} &= T^{start}_{\text{free}} + D_{\text{free}}; \\ T^{end}_{\text{sensing}} &= T^{start}_{\text{sensing}} + D_{\text{sensing}}; \\ T_{\text{start}} &\le T^{start}_{\text{free}},\ T^{start}_{\text{sensing}}; \\ T^{end}_{\text{free}},\ T^{end}_{\text{sensing}} &\le T_{\text{end}}; \\ (T^{end}_{\text{free}} \le T^{start}_{\text{sensing}}) \quad &\text{or} \quad (T^{end}_{\text{sensing}} \le T^{start}_{\text{free}}). \end{aligned} \tag{3}$$

(2) Causality and QoS Constraints: Each WiFi FG must respect its arrival time and deadline. If scheduled within the current cycle ($P_k = 0$), it must satisfy the arrival constraint (i.e., $s'_k \ge a_k$) and complete before both its deadline and the cycle boundary. Otherwise, if postponed ($P_k = 1$), it must remain schedulable in the next cycle:

$$\begin{aligned} a_k &\le s'_k, && (\text{if } P_k = 0); \\ s'_k + d_k &\le \min(e_k, T_{\text{end}}), && (\text{if } P_k = 0); \\ T_{\text{end}} + d_k &\le e_k, && (\text{if } P_k = 1). \end{aligned} \tag{4}$$

(3) Inter-FG Conflict Avoidance: To prevent internal WiFi collisions, scheduled FGs within the same cycle must not overlap in time:

$$\begin{aligned} &(s'_k + d_k \le s'_j) \quad \text{or} \quad (s'_j + d_j \le s'_k) \quad \forall k \ne j, \\ &\text{if } P_k, P_j = 0. \end{aligned} \tag{5}$$

The above formulation defines a constrained scheduling problem over a finite time horizon. Due to the presence of indicator functions and logical conditions, the original problem is non-linear. We reformulate it into a Mixed-Integer Linear Programming (MILP) problem using standard linearization techniques (e.g., the Big-M method), with detailed derivations provided in Appendix A.

In practice, the number of FGs within each scheduling cycle remains limited due to traffic aggregation, which significantly reduces the problem scale compared to packet-level scheduling. As a result, the MILP can be solved using off-the-shelf solvers at the AP, enabling practical deployment within typical scheduling intervals.

To further reduce computational overhead in resource-constrained scenarios, we also consider a lightweight scheduling alternative. Instead of explicitly solving the optimization problem, the AP exploits existing whitespace in the WiFi traffic schedule. If sufficient idle intervals are available, the required interference-free window for ZigBee can be directly allocated. Otherwise, the AP extends the largest available whitespace as much as possible and assigns the resulting interval to the ZigBee Coordinator.

Although this approach does not guarantee optimality, it preserves the core design principle of CoWiC-MAC — creating flexible transmission opportunities for ZigBee with minimal disruption to WiFi traffic — while significantly reducing computational complexity. Importantly, the residual WiFi activity within the reserved window, resulting from FG-level rescheduling decisions, is explicitly exposed to the ZigBee network as fine-grained temporal fragments, which form the basis for the fragment-aware adaptation described in Section 4.4.

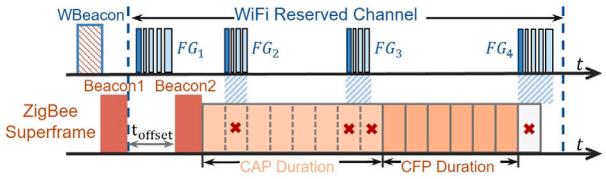


Fig. 6. Schematic of ZigBee adaptive transmission.

4.4. ZigBee adaptive transmission

However, due to the inherent trade-off between WiFi QoS and ZigBee access, the interference-free window reserved by the AP may still contain residual WiFi frame groups (FGs). In practical heterogeneous environments, reserving a fully interference-free interval is therefore often difficult. To address this, the WiFi AP provides the ZigBee Coordinator (ZC) with a fine-grained temporal map of channel occupancy via signaling frames (e.g., WBeacons). This map specifies both the reserved interval and the timing of unavoidable residual WiFi FGs within it. Based on this information, the ZC invokes the Lightweight Window-Scaling Algorithm (L-WSA), which dynamically adapts the ZigBee superframe by adjusting the timeslot duration (t_{SfSlot}) and the CAP/CFP partition, enabling ZigBee transmissions to effectively navigate fragmented WiFi interference.

As illustrated in Fig. 6, the L-WSA algorithm adapts the ZigBee superframe to fragmented WiFi activity. The beacon offset t_{offset} is first adjusted to avoid early interference. Within the active period, the ZC evaluates the impact of unavoidable WiFi fragments (i.e., residual WiFi FGs) on each timeslot and selectively utilizes only interference-free intervals. The adaptation is asymmetric: minor overlaps (e.g., within CAP) may be tolerated with limited impact, whereas critical overlaps (e.g., near CFP boundaries) are proactively avoided to prevent inefficient transmissions. This fragment-aware adjustment enables efficient spectrum utilization while maintaining protocol stability under cross-technology interference.

The model focuses on selecting superframe parameters that maximize effective transmission under fragmented channel availability. To operationalize the adaptive strategies described above, we establish a formal optimization model. The parameters and decision variables involved in the ZigBee adaptive transmission are summarized in Table 3.

(1) Optimization Objective and Efficiency Metrics: The L-WSA algorithm seeks to maximize the total transmission efficiency η_{total} within a single superframe by dynamically partitioning the available airtime. As formulated in Eq. (6), the objective function balances the throughput gains from the CAP and the CFP, where a weight factor α is introduced to prioritize the deterministic nature of CFP transmissions.

$$\max_{t_{\text{SfSlot}}, i, j, t_{\text{offset}}} \eta_{\text{total}} = m \cdot t_{\text{min}} \cdot \eta^{p}_{\text{cap}} + \alpha \cdot n \cdot t_{\text{SfSlot}} \cdot \eta^{p}_{\text{cfp}}; \tag{6}$$

The per-packet efficiencies, $\eta^{\text{p}}_{\text{cap}}$ and $\eta^{\text{p}}_{\text{cfp}}$, quantify the ratio of effective payload to total transmission time, where t_{ovh} accounts for protocol overhead (e.g., backoff, ACK, turnaround):

$$\eta^{\text{p}}_{\text{cap}} = \frac{t_{\text{pk}} - t_{\text{ovh}}}{t_{\text{min}}}, \quad \eta^{\text{p}}_{\text{cfp}} = \frac{t_{\text{pk}} - t_{\text{ovh}}}{t_{\text{SfSlot}}}. \tag{7}$$

(2) Physical Constraints and Superframe Mapping: To maintain standard compatibility, the packet transmission time t_{pk} is adaptively bounded by the slot duration t_{SfSlot} and a hardware-defined threshold. This ensures that each transmission, including backoff, turnaround (t_{ta}), and acknowledgment (t_{ack}), remains within slot boundaries:

$$\begin{aligned} t_{\text{pk}} &= \min\left(t_{\text{SfSlot}},\ t^{\max}_{\text{pk}} + t_{\text{ta}} + t_{\text{ack}}\right) - t_{\text{ta}} - t_{\text{ack}}; \\ t_{\text{min}} &= 2 \cdot t_{\text{BoSlot}} + t_{\text{pk}}. \end{aligned} \tag{8}$$

Table 3
Variables and definitions in ZigBee traffic adjustment.

Variables	Definitions	Variables	Definitions
t_{SfSlot}	Superframe timeslot duration	T_{beacon}	ZigBee beacon interval
t_{BoSlot}	Backoff slot duration in slotted CSMA/CA	T_{CAP}	CAP duration
i	Number of CAP timeslots	T_{CFP}	CFP duration
j	Number of CFP timeslots	m	Number of valid CAP transmissions
t_{offset}	Beacon offset relative to WiFi idle period	n	Number of valid CFP transmissions
t_{ta}	Turnaround time defined in IEEE 802.15.4	η^p_{cap}	Per-packet efficiency in CAP
t_{ack}	ACK response time	η^p_{cfp}	Per-packet efficiency in CFP
t_{pk}	ZigBee packet transmission time	α	Weight factor for CFP efficiency
t^{th}_{slot}	Threshold for superframe timeslot duration	η_{total}	Total transmission efficiency
t_{min}	Minimum CAP transmission interval	dur_{start}	Start time of WiFi idle period
		dur_{end}	End time of WiFi idle period

Based on the beacon offset t_{offset}, the temporal boundaries of the superframe ($T_{\text{beacon}}, T_{\text{CAP}}, T_{\text{CFP}}$) are linearly mapped within the granted transmission interval $[dur_{\text{start}}, dur_{\text{end}}]$. This mapping enables the ZC to shift the superframe to avoid early interference from WiFi FGs, such as FG_1 in Fig. 6.

(3) Fragment-Aware Slot Selection: L-WSA explicitly accounts for the set of unavoidable WiFi fragments $\mathcal{T}^{\text{WiFi}}$ (i.e., residual WiFi FGs within the reserved window). The number of valid transmissions (m, n) is computed by selecting only those slots that do not overlap with WiFi activity:

$$\begin{aligned} m &= \sum_{k\in\text{CAP}} \mathbb{1}\left([t_k,\ t_k + t_{\min}] \cap \mathcal{T}^{\text{WiFi}} = \emptyset\right); \\ n &= \sum_{l\in\text{CFP}} \mathbb{1}\left([t_l,\ t_l + t_{\text{SfSlot}}] \cap \mathcal{T}^{\text{WiFi}} = \emptyset\right). \end{aligned} \tag{9}$$

Crucially, the total span of the active period, governed by the indices i and j, is strictly constrained by the granted whitespace duration $(dur_{\text{end}} - dur_{\text{start}})$ and the standard 16-slot limit:

$$(1 + i + j) \cdot t_{SfSlot} \le dur_{end} - dur_{start}, \quad 0 \le j \le 7,\ i + j \le 15. \tag{10}$$

Although the L-WSA formulation is a Mixed-Integer Nonlinear Programming (MINLP) problem, its complexity is tightly bounded by the standard-defined parameter space. Specifically, t_{SfSlot} is selected from a discrete set, and (i, j) lie within small finite ranges. As a result, the optimization reduces to a low-dimensional discrete search problem, which can be efficiently solved via enumeration or lookup tables, enabling real-time execution on resource-constrained ZigBee devices.

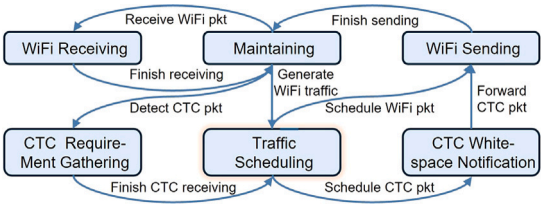
4.5. Implementation considerations

This section discusses practical considerations for implementing CoWiC-MAC in real-world deployments. We first present the coordinated execution logic of the protocol using the Finite State Machines (FSM), providing a system-level view of cross-technology signaling, WiFi scheduling, and ZigBee adaptation. We then discuss key deployment challenges, including CTC robustness, protocol compatibility, and scalability in multi-AP scenarios.

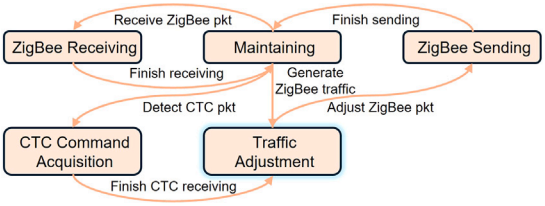
4.5.1. System-level coordination via FSM

To provide a system-level view of how the proposed mechanisms are executed in practice, we define the FSM for both the WiFi AP and the ZC, as illustrated in Fig. 7. These FSM models formalize the operational logic of the four-step cross-technology handshake and bridge the signaling, scheduling, and adaptation modules into a closed-loop coordination process.

WiFi AP Logic: The AP operates around its primary communication loop between the Maintaining and WiFi Sending/Receiving states. Upon detecting an uplink CTC packet, it transitions to the CTC Requirement Gathering state to extract ZigBee communication demands. These demands, together with the locally observed WiFi traffic status, are fed into the traffic scheduling module. The AP then enters the CTC White-space Notification state to broadcast the computed coordination parameters (via WBeacon), before executing the scheduled transmissions.



(a) State Machine Diagram of the WiFi AP



(b) State Machine Diagram of the ZigBee Coordinator

Fig. 7. Coordinated Finite State Machine Design in the CoWiC-MAC Protocol.

ZigBee Coordinator Logic: The ZC monitors for downlink cross-technology signals in the CTC Command Acquisition state. Once a WBeacon is successfully decoded, it transitions to the Traffic Adjustment state, where it reshapes the ZigBee superframe and schedules transmissions to align with the allocated interference-free windows.

This FSM-based design enables tightly coordinated interaction between the two networks while maintaining operational robustness. In the event of CTC handshake failure, both entities revert to their default CSMA/CA-based operation, ensuring backward compatibility and uninterrupted communication.

4.5.2. CTC robustness and synchronization

Despite recent advances in CTC, practical deployments inevitably face packet loss, decoding errors, and timing uncertainties due to interference and hardware limitations. CoWiC-MAC explicitly accounts for these imperfections. CTC is primarily used for exchanging control information rather than continuous data transmission, which reduces sensitivity to occasional errors. In practice, CTC failures manifest as missed or corrupted ZBeacon/WBeacon exchanges. If the WiFi AP does not receive a valid demand report, it assumes no coexistence requirement and maintains its default transmission behavior. If the ZC fails to decode a WBeacon, it remains in its default channel-listening state and reattempts coordination in subsequent cycles.

To further enhance robustness, CoWiC-MAC adopts a periodic coordination mechanism in which ZigBee demand reports and WiFi responses are refreshed every scheduling cycle, enabling natural recovery from transient CTC failures without complex retransmission schemes.

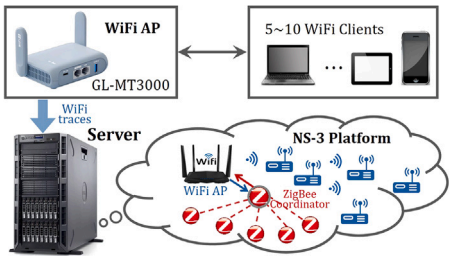


Fig. 8. Experimental platform for CoWiC-MAC.

From a synchronization perspective, CoWiC-MAC does not require strict global time alignment. Instead, WBeacons convey beacon interval information and the relative positions of interference-free and sensing windows, allowing the ZigBee network to align its activity within designated intervals without tight synchronization. This relaxed requirement simplifies implementation and improves robustness against clock drift.

Overall, even if CTC signaling fails in a given cycle, both the AP and the ZC revert to their default CSMA/CA-based operation. Therefore, CTC imperfections degrade coordination efficiency but do not compromise baseline network functionality.

4.5.3. Protocol compatibility and deployment considerations

CoWiC-MAC is designed as a lightweight overlay that operates without modifying the underlying IEEE 802.11 and IEEE 802.15.4 standards. All coordination is realized through existing frame exchanges, enabling incremental deployment with minimal protocol stack modifications. Moreover, all CoWiC functionalities are optional rather than mandatory, ensuring seamless coexistence with legacy devices. The proposed design is compatible with mainstream IEEE 802.11 ax/be WiFi systems and IEEE 802.15.4-based ZigBee networks.

Importantly, CoWiC-MAC does not rely on specific physical-layer features such as MU-MIMO or directional beamforming, although such capabilities can further improve spatial reuse and mitigate residual interference. Since the coordination operates at the MAC layer, it remains applicable regardless of whether WiFi transmissions are single-user or multi-user, as long as transmission timing can be controlled at the AP.

From a deployment perspective, the core functionalities of CoWiC-MAC—including WiFi-side traffic rescheduling and ZigBee-side adaptive superframe configuration—are executed locally at the AP and the ZC, respectively. No centralized controller or cross-network infrastructure is required. This distributed design simplifies integration and reduces system overhead, making the protocol practical for real-world heterogeneous environments.

In multi-AP scenarios, inter-AP interference may affect the stability of the reserved interference-free windows. CoWiC-MAC adopts a single-AP operational paradigm, where each AP independently performs coordination based on locally observed traffic. ZigBee activity is treated as a special class of managed traffic within the AP's scheduling domain, and ZigBee-related reservations are naturally embedded into existing WiFi coordination mechanisms, such as channel access, channel selection, and spatial reuse. As a result, CoWiC-MAC does not introduce new forms of inter-AP contention beyond those already present in conventional WiFi deployments. While dense multi-AP environments may impact the stability of reserved windows, such effects primarily influence coordination efficiency rather than system correctness. Extending the framework to support explicit inter-AP coordination remains an important direction for future work.

Overall, CoWiC-MAC maintains compatibility with existing standards while enabling practical deployment with low implementation overhead. At the same time, it ensures robustness against CTC imperfections and provides a clear path toward scalability in more complex and dynamic network environments.

5. Evaluation

In this section, we conduct a comprehensive evaluation of the CoWiC-MAC protocol by integrating real-world WiFi traffic traces with extensive NS-3 simulations, systematically analyzing its performance across various traffic intensities and coordination configurations to validate its effectiveness in heterogeneous coexisting environments.

5.1. Experiment settings

To evaluate the performance of CoWiC-MAC, we conducted extensive simulations on the NS-3 platform [41], utilizing a trace-driven methodology to bridge the gap between theoretical modeling and real-world conditions. As illustrated in Fig. 8, the experimental platform consists of two main stages: data collection and network simulation. In the first stage, a GL.iNet GL-MT3000 router serves as the WiFi AP, catering to 5–10 WiFi clients (including smartphones and laptops) in typical office scenarios. During data collection, these clients engaged in diverse and concurrent network activities such as web browsing, video conferencing, online gaming, and HD streaming. We collected four 10-minute traces of this mixed and stochastic traffic using Tcpdump tools, each exhibiting distinct statistical characteristics, which were then fed into a central server.

Based on these traces, WiFi traffic is further aggregated into Frame Groups (FGs) according to temporal proximity and scheduling relevance, forming the basic unit for rescheduling in our system. To accommodate differentiated QoS requirements, we set the maximum scheduling postponement bounds for different WiFi FG categories to 50 ms, 100 ms, 150 ms, and 200 ms, respectively.

To validate the CoWiC mechanism at the MAC layer, a USRP-based hardware implementation was deemed infeasible due to the substantial and unpredictable communication latencies between the USRP front-end and its PC host [25], which would obscure the precise timing required for microsecond-level coordination. Consequently, we implemented the protocol logic within an NS-3 environment running on the server, where WiFi network activity is driven by replayed real-world traces at the FG level. The simulated ZigBee network comprises one ZigBee Coordinator (ZC) and five end devices in a star topology, operating within the temporal windows dynamically allocated by the WiFi AP based on coexistence requirements. To maintain a rigorous evaluation, we assume that any temporal overlap between WiFi FGs and ZigBee transmissions on the same channel results in mutual corruption, rendering both transmissions unsuccessful.

For the baseline configurations, we implemented six experimental groups (Exp 1–6) to systematically benchmark our work against both standard protocols and state-of-the-art coordination schemes. As summarized in Table 4, Exp 1 serves as the baseline for legacy coexistence without any coordination. Exp 2 represents an opportunistic whitespace detection approach, where the WiFi AP identifies the largest contiguous gap within a cycle and notifies the ZigBee node via unidirectional CTC, without rescheduling WiFi FGs or adapting ZigBee transmission. Exp 3 implements the existing coordination (ECC [22]) scheme, featuring bidirectional CTC but restricted to static ZigBee transmission. Exp 4 represents the full implementation of CoWiC-MAC, integrating MILP-based soft-constraint FG scheduling with L-WSA adaptive transmission. A mandatory 10 ms sensing window is incorporated to ensure closed-loop coordination, representing the most robust configuration of our framework.

To further isolate the contributions of individual modules through ablation studies, Exp 5 adopts a strict "WiFi-Vacate" strategy, where all WiFi FGs overlapping with the reserved window are forcibly removed to guarantee a clean ZigBee interval. Exp 6 evaluates best-effort FG rescheduling combined with adaptive L-WSA transmission in the absence of a sensing window.

To quantify the efficiency and impact of these strategies, six key performance metrics are defined. First, the ZigBee Transmission Gain

Table 4
Configuration of Experimental Groups and Comparative Baselines.

Exp. ID	WiFi Scheduling	CTC Direction	ZigBee Adaptation	Sensing Window	Goal/Category
Exp 1	None (Static)	None	Fixed	None	Baseline (Static)
Exp 2	None (Gap Search)	WiFi → ZigBee	Fixed	None	Baseline (Opportunistic)
Exp 3	ECC-based	Bidirectional	Fixed	None	ECC Scheme
Exp 4	MILP (Soft Const.)	Bidirectional	L-WSA	10 ms (Default)	**CoWiC-MAC (Full)**
Exp 5	MILP (Hard Const.)	WiFi → ZigBee	Fixed	None	Ablation: Rigid Scheduling.
Exp 6	MILP (Soft Const.)	WiFi → ZigBee	L-WSA	None	Ablation: Overhead-free

Ratio (G_z) is calculated as the ratio of the actual successfully delivered ZigBee payload (i.e., transmissions that avoid collisions within the allocated D_{free} window) to the theoretical maximum payload under ideal interference-free conditions. This metric reflects the system's ability to utilize reserved whitespaces effectively. Second, the WiFi Frame Group Drop Rate (PDR_w) measures the proportion of WiFi FG transmission duration that is forced to be discarded due to scheduling constraints, serving as a direct indicator of the impact on primary network reliability. Third, the Effective Spectrum Utilization (U_{eff}) represents the fraction of total time effectively occupied by successful WiFi transmissions, ZigBee activity, and sensing overhead. Finally, to capture the delay characteristics and scheduling overhead introduced by coordination, we jointly consider three related metrics: the WiFi latency jitter, the number of moved WiFi FGs, and the maximum FG delay. Together, these metrics characterize the temporal variation, the extent of scheduling intervention, and the worst-case delay impact on WiFi traffic after rescheduling.

5.2. WiFi traffic analysis

Before evaluating the system performance, we first analyze the statistical characteristics of the collected WiFi traffic traces to highlight the challenges of coexistence. Fig. 9 illustrates the Complementary Cumulative Distribution Function (CCDF) of the natural whitespace lengths across four representative traffic groups. It is observed that as WiFi traffic intensity increases, the availability of contiguous transmission windows for ZigBee diminishes drastically. For instance, in Traffic I and II, the probability of encountering a whitespace longer than 10 ms is 79.2% and 65.9%, respectively. However, in the high-intensity Traffic IV, this probability plummets to a mere 4.1%, effectively precluding stable communication for legacy ZigBee devices.

Quantitative insights from the traffic traces are summarized in Table 5. To provide a granular view of the traffic composition, we categorize the WiFi Frame Groups (FGs) based on their original QoS requirements: *QoS-Sensitive FGs* are defined by a stringent delay bound (≤50 ms, e.g., real-time video or commands), while *Non-real-time FGs* represent delay-tolerant applications (>100 ms, e.g., web browsing). The results reveal that as the network transitions from sparse to saturation (Traffic I to IV), the *QoS-Sensitive FG Ratio* significantly rises from 7.3% to 45.2%, while the *Non-real-time FG Ratio* drops from 84.9% to 39.7%. Correspondingly, the *Avg. Delay Bound* decreases from 160.6 ms to 102.1 ms. This trend indicates that under high-load conditions, the WiFi AP faces a "tightening" scheduling window: it must not only manage higher traffic volume but also satisfy more stringent latency constraints for a larger proportion of traffic. Furthermore, the *Avg. Natural Whitespace* (ranging from 5.3 to 142.3 ms) varies drastically across different groups, indicating that "usable" gaps and "fragmented" micro-slots may emerge dynamically at different stages. Such temporal instability cannot guarantee the reliable transmission of a standard ZigBee frame (typically 1–4 ms), validating that opportunistic access is fundamentally insufficient and necessitating the active channel reservation proposed in Section 4.3.

Furthermore, Fig. 10 depicts the temporal distribution of WiFi data volume over a 600-second interval. The results reveal a high degree of burstiness and stochastic fluctuation. This temporal volatility, combined with the fragmented nature of available airtime, underscores the necessity of the coordination synergy in CoWiC-MAC.

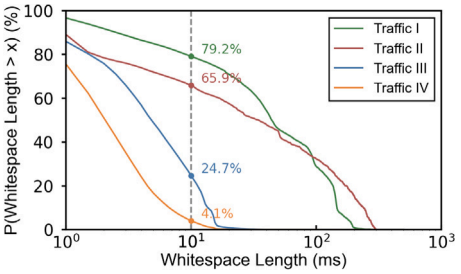


Fig. 9. CCDF of whitespace lengths for different WiFi traffic groups.

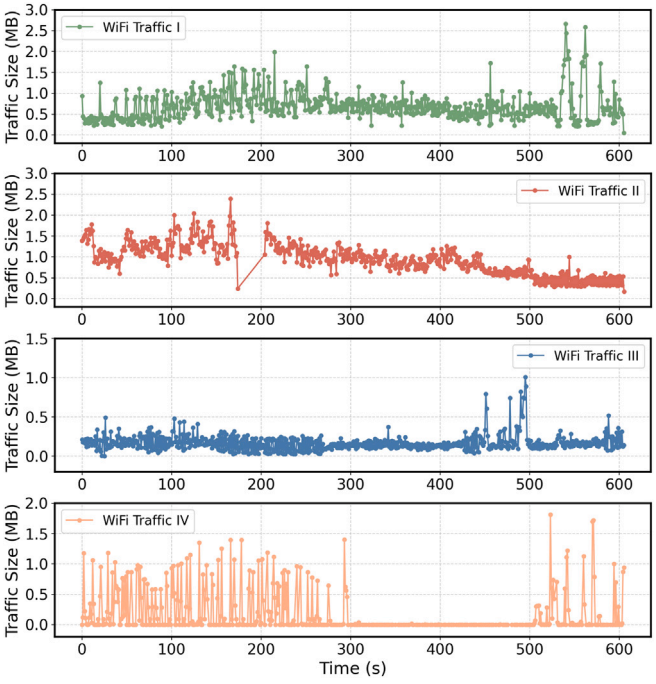


Fig. 10. Distribution of data size per second in different WiFi traffic groups.

Specifically, the WiFi AP utilizes MILP-based scheduling to proactively create interference-free windows by navigating through these bursts, while the ZC employs the L-WSA algorithm to "harvest" residual temporal fragments in real-time. This joint optimization ensures that the system can effectively adapt to fluctuating traffic patterns without compromising the QoS of the primary WiFi service.

5.3. Performance of CoWiC-MAC

5.3.1. Performance benchmark in high-load scenarios

We first conduct a comprehensive analysis of Fig. 11 (Traffic I, T_{ZB} = 1000 ms), which serves as the representative baseline for our evaluation. For clarity of presentation, the horizontal axis corresponds to different D_{free} configurations, while the vertical axes represent the six performance metrics.

Table 5
Statistical Characteristics of Representative WiFi Traffic Traces Aggregated by Frame Groups.

Statistics	Traffic I	Traffic II	Traffic III	Traffic IV
Avg. FG Duration (ms)	39.2	23.7	16.0	25.8
Avg. Natural Whitespace (ms)	5.3	9.2	34.3	142.3
QoS-Sensitive FG Ratio (%)	7.3	10.7	27.4	45.2
Non-real-time FG Ratio (%)	84.9	70.5	49.2	39.7
Avg. Delay Bound (ms)	160.6	146.2	120.7	102.1

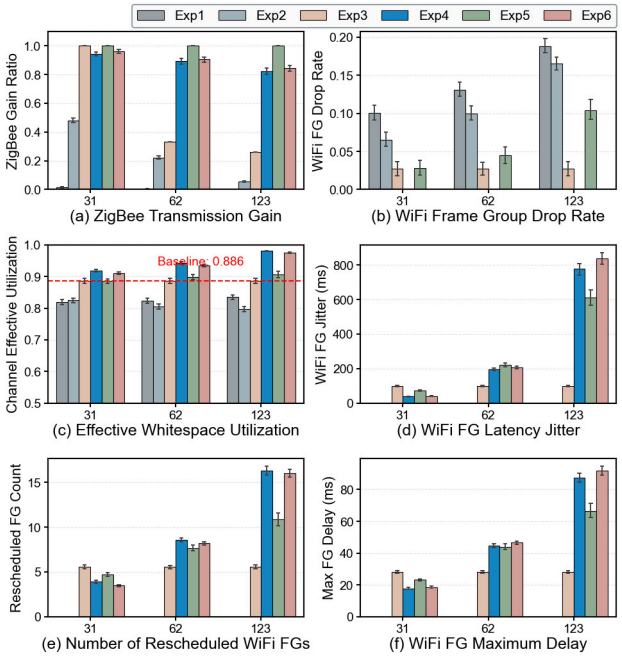


Fig. 11. Performance Evaluation of CoWiC-MAC under High-Load WiFi Traffic (Traffic I) with $T_{ZB} = 1000$ ms.

Legacy and Opportunistic Approaches (Exp 1 & 2): For the uncoordinated Exp 1, the ZigBee Transmission Gain Ratio (G_z) is a negligible 1.3% at $D_{free} = 31$ ms and drops nearly to zero as D_{free} increases. Severe interference leads to a WiFi Frame Group Drop Rate (PDR_w) of 9.7% at $D_{free} = 123$ ms, resulting in an Effective Spectrum Utilization (U_{eff}) well below the baseline. This confirms that uncoordinated transmission is nearly infeasible under dense WiFi traffic.

For Exp 2, although unidirectional CTC provides an initial G_z of 55.8% at $D_{free} = 31$ ms, this gain sharply decays to 8.5% as D_{free} reaches 123 ms, while PDR_w increases to 7.6%. Similar to Exp 1, persistent collisions suppress overall U_{eff}. These results indicate that relying solely on inherent whitespaces is insufficient to guarantee reliable coexistence, even with basic cross-technology signaling.

Static and Rigid Coordination (Exp 3 & 5): At $D_{free} = 31$ ms, Exp 3 and Exp 5 exhibit similar performance, both achieving near-complete ZigBee transmission with less than 1% WiFi FG loss. However, Exp 5 yields lower WiFi latency jitter due to its flexible window placement, whereas Exp 3 is constrained by fixed reservation positions.

A clear divergence appears as D_{free} increases. Due to the inherent limitation of the legacy WiFi NAV mechanism, Exp 3 cannot reserve windows beyond 40 ms, leading to performance stagnation at larger D_{free}. In contrast, Exp 5 enforces rigid reservations to fully satisfy ZigBee demand, maintaining $G_z = 100\%$. This comes at a cost: at $D_{free} = 123$ ms, PDR_w reaches 2.8%, indicating non-negligible degradation of WiFi service.

Proposed CoWiC Mechanism (Exp 4 & 6): The proposed CoWiC-MAC (Exp 4 and Exp 6) achieves the most balanced and robust performance. At $D_{free} = 31$ ms, both configurations achieve $G_z > 98.6\%$. As demand increases to $D_{free} = 62$ ms, the gain remains high at 96.3%, demonstrating strong adaptability under heavy WiFi load. Even at $D_{free} = 123$ ms, G_z remains stable between 93.4% and 94.7%, while consistently achieving zero WiFi FG loss—an outcome not observed in any baseline scheme. This improvement is enabled by the MILP-based postponement mechanism, which flexibly reshapes WiFi transmissions instead of enforcing rigid reservations.

The difference between Exp 4 and Exp 6 reflects a design trade-off: the dedicated sensing window in Exp 4 slightly reduces G_z due to temporal overhead, but improves U_{eff} and enhances long-term coordination stability.

In addition to latency jitter, we further examine the number of rescheduled (moved) WiFi FGs and the maximum FG delay. These three metrics consistently characterize the impact of scheduling adjustments introduced by CoWiC-MAC. Exp 1 and Exp 2 exhibit near-zero jitter and no rescheduling, as no coordination is performed. In contrast, Exp 3 and Exp 5 show higher jitter at small D_{free} due to rigid scheduling constraints. For CoWiC-MAC (Exp 4 & 6), moderate increases in jitter, moved FG count, and maximum delay are observed as D_{free} grows. However, these increases remain well-bounded and scale smoothly with ZigBee demand. This indicates that CoWiC-MAC introduces controlled and predictable scheduling overhead, affecting only a limited subset of WiFi FGs while preserving overall QoS stability.

5.3.2. Generalizability and sensitivity analysis

Impact of ZigBee Periods: Building upon the baseline analysis, we examine the system's generalizability across extended periods. As T_{ZB} increases to 2000 ms (Fig. 12) and 5000 ms (Fig. 13), the G_z of Exp 1 remains near zero, confirming that uncoordinated transmission is ineffective under dense WiFi traffic. The G_z of Exp 2 shows a gradual increase with larger T_{ZB}, indicating that a longer temporal horizon improves the probability of finding usable whitespaces. For the coordinated schemes (Exp 3–6), all maintain near-optimal G_z (approaching 1.0), except for the NAV-limited behavior in Exp 3.

As T_{ZB} increases, the growing availability of natural whitespaces leads to convergence in U_{eff} across all schemes. Nevertheless, Exp 4 consistently achieves the highest utilization, demonstrating its ability to minimize idle airtime even under relaxed temporal constraints. Meanwhile, delay-related metrics (jitter, moved FGs, and maximum delay) exhibit consistent decreasing trends with increasing T_{ZB}, reflecting the reduced scheduling pressure and diminished need for traffic rescheduling.

Impact of Traffic Loads: A comparison between Fig. 11 and Fig. 14 highlights the impact of WiFi traffic intensity. In sparse scenarios (Traffic III), even the uncoordinated Exp 1 achieves a G_z of 54.3% at $D_{free} = 31$ ms. Despite the increase of D_{free} to 123 ms, Exp 1 still sustains a G_z of 36.1%, which is significantly higher than its performance in dense traffic, underscoring that traffic density is a key factor for stochastic ZigBee access. Under these sparse conditions, Exp 2 achieves a G_z of 93.9% with only 1.4% WiFi frame group drop rate at $D_{free} = 123$ ms, indicating that unidirectional CTC-based coordination can be effective when channel contention is low. Nevertheless, the consistent performance of Exp 4 across all traffic loads further validates the robustness of the CoWiC mechanism.

Regarding delay-related metrics (jitter, moved FGs, and maximum delay), a clear distinction is observed between high-load and low-load scenarios. Under high WiFi traffic, these metrics are noticeably higher due to increased scheduling pressure, yet they remain within acceptable

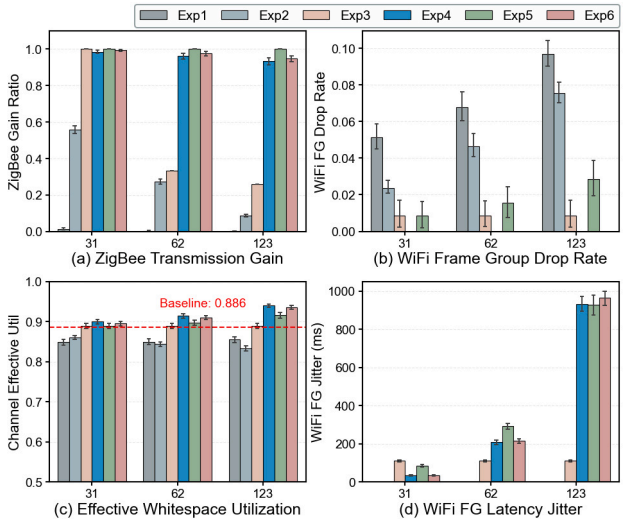


Fig. 12. Performance Evaluation of CoWiC-MAC under High-Load WiFi Traffic (Traffic I) with $T_{ZB} = 2000$ ms.

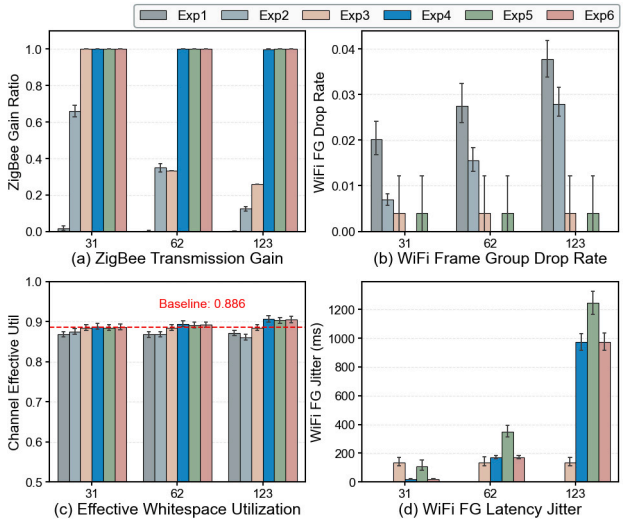


Fig. 13. Performance Evaluation of CoWiC-MAC under High-Load WiFi Traffic (Traffic I) with $T_{ZB} = 5000$ ms..

bounds. In contrast, under low-load conditions, Exp 4–6 generally exhibit lower delay-related metrics compared to Exp 3, indicating that the cost of rescheduling WiFi frame groups is reduced when sufficient natural whitespaces are available.

To further evaluate the sensitivity to WiFi loads, we conduct evaluations under $T_{ZB} = 1000$ ms and $D_{free} = 63$ ms across the full range of WiFi datasets, as shown in Fig. 15. When WiFi channel occupancy is below 0.6, Exp 2, 4, 5, and 6 all achieve near-optimal G_z with negligible WiFi frame group drop rate (PDR_w). However, as occupancy exceeds 0.6, the G_z for most schemes begins to decline, while the PDR_w for Exp 2 and 5 increases significantly. This divergence highlights the superiority of the CoWiC mechanism in balancing bi-network communication under heavy loads. Furthermore, U_{eff} and Jitter exhibit trends consistent with WiFi load, validating the physical realism of the proposed scheduling model.

Sensitivity of the Sensing Window ($D_{sensing}$): Finally, we investigate the impact of the sensing window size (Exp 4-1 to 4-4, with $D_{sensing}$ set to 10, 20, 50, and 100 ms, respectively) at $T_{ZB} = 1000$ ms, as depicted in Fig. 16 and Fig. 17. In dense traffic, G_z exhibits a monotonic

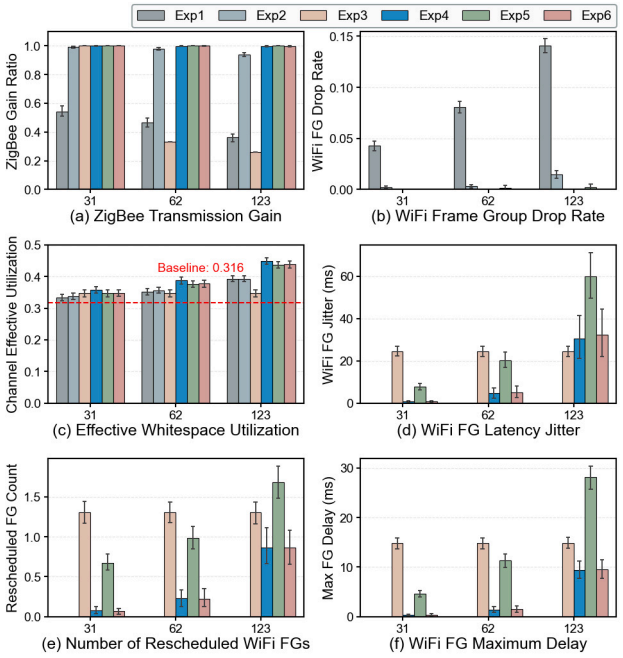


Fig. 14. Performance Evaluation of CoWiC-MAC under Low-Load WiFi Traffic (Traffic III) with $T_{ZB} = 1000$ ms.

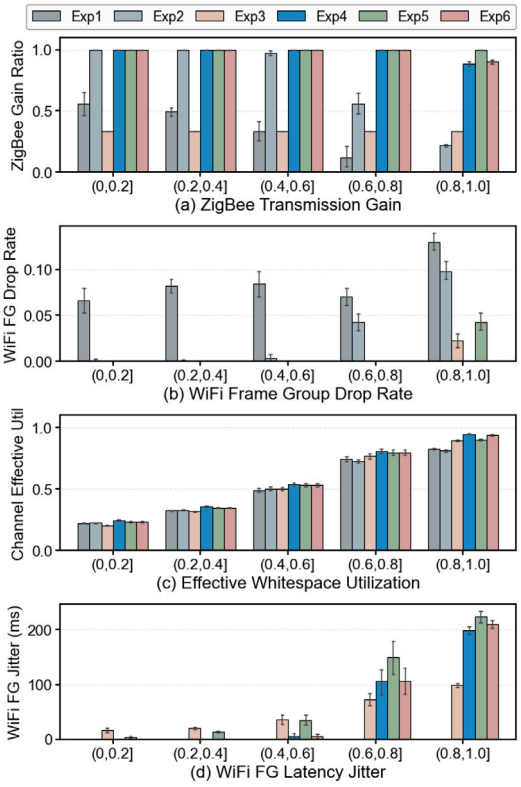


Fig. 15. Performance Evaluation of CoWiC-MAC under Varying WiFi Channel Occupancy Levels.

decrease as $D_{sensing}$ increases, dropping from 90.5% (10 ms) to 48.0% (100 ms). When the ZigBee period is short and traffic is dense, an oversized sensing window (the feedback link from ZigBee to WiFi) encroaches significantly upon the ZigBee transmission time and induces substantial WiFi delay-related overhead (e.g., jitter). Conversely, in

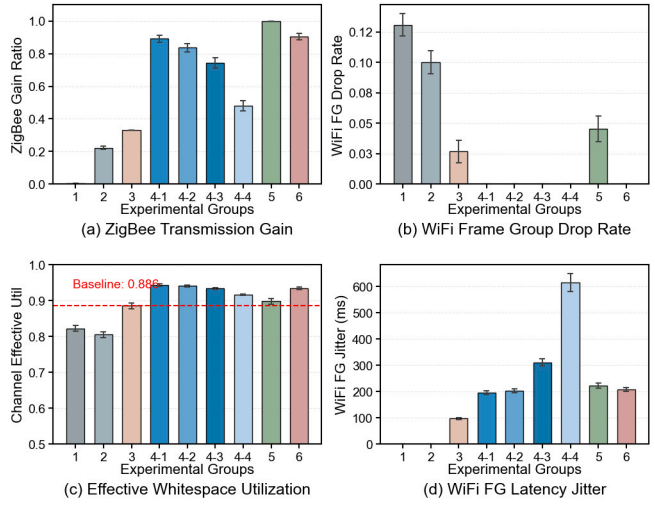


Fig. 16. Performance Evaluation of CoWiC-MAC with Various $D_{sensing}$ Settings under WiFi Traffic I.

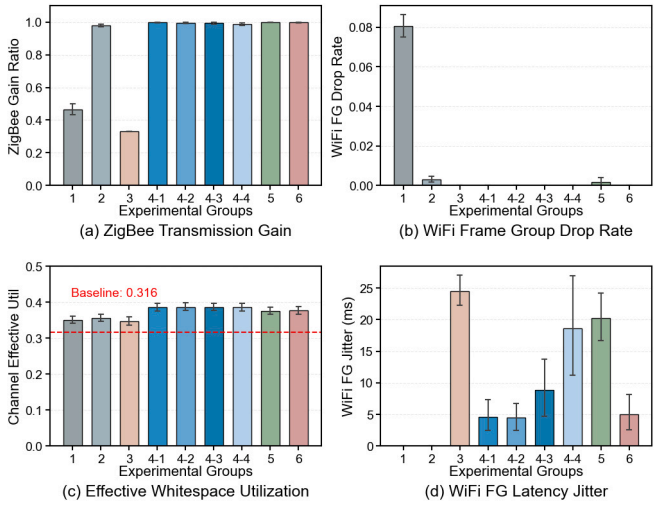


Fig. 17. Performance Evaluation of CoWiC-MAC with Various $D_{sensing}$ Settings under WiFi Traffic III.

sparse traffic, all CoWiC-enabled configurations perform well, and the choice of $D_{sensing}$ primarily affects delay-related metrics rather than throughput-related performance. This confirms that a compact sensing window is essential for efficient coexistence in congested spectrum environments.

In summary, the experimental results across diverse traffic scenarios and parameter settings consistently demonstrate the effectiveness of CoWiC-MAC. By synergizing MILP-based proactive scheduling with the L-WSA adaptive transmission mechanism, the proposed protocol significantly improves the throughput of the ZigBee network while maintaining the reliability of the WiFi service with near zero frame group loss and bounded delay overhead. These results confirm that CoWiC-MAC achieves a favorable balance between spectrum efficiency and cross-technology coexistence, providing a practical and scalable solution for heterogeneous wireless environments.

6. Conclusion

In this study, we investigated the stochastic interference challenges between primary WiFi and secondary ZigBee networks and proposed CoWiC-MAC as a bidirectional coordination framework. The core of our contribution lies in shifting from a passive, contention-based paradigm to a proactive and adaptive scheduling approach. Our design integrates a MILP-based traffic reordering mechanism at the WiFi AP with a lightweight L-WSA adaptive transmission scheme at the ZigBee coordinator, bridged by a cross-technology coordination link. This synergistic design demonstrates that coexistence can be effectively improved through strategic traffic reshaping rather than rigid resource reservation.

Our trace-driven evaluations show that CoWiC-MAC effectively decouples ZigBee performance from WiFi traffic fluctuations. Under near-saturation conditions (approximately 90% channel occupancy), the protocol achieves a ZigBee gain ratio of up to 96.3% while maintaining zero WiFi frame group loss in our evaluated scenarios—performance that is difficult to achieve with legacy uncoordinated or unidirectional schemes. Overall, CoWiC-MAC achieves a favorable balance between WiFi service reliability and ZigBee transmission efficiency.

Looking forward, the coordination granularity can be further enhanced by extending the scheduling framework toward multi-dimensional time-frequency resource orchestration. In particular, leveraging OFDMA capabilities in next-generation WiFi standards (e.g., IEEE 802.11ax/be) provides opportunities for finer-grained resource allocation. Beyond these technical extensions, the proposed framework can be generalized to broader multi-technology coexistence scenarios. By incorporating additional protocols such as BLE and LoRa, it can support cross-technology coordination in increasingly complex IoT environments, moving toward more adaptive and spectrum-efficient wireless systems.

CRediT authorship contribution statement

Tao Cheng: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Shining Li:** Project administration, Funding acquisition. **Yan Pan:** Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. MILP reformulation of WiFi traffic rescheduling

Directly optimizing the objective in Eq. (1) is computationally challenging due to the presence of non-linear indicator functions and logical conditions. To address this, we reformulate the problem into a Mixed-Integer Linear Programming (MILP) framework using standard linearization techniques, particularly the Big-M method. This transformation converts conditional logic into a set of linear constraints, enabling efficient solution via off-the-shelf solvers.

Additional Variables for MILP Reformulation: In addition to the variables defined in Table 2, the MILP formulation introduces the following auxiliary variables:

- $z_k^{\text{free}} \in \{0,1\}$: Indicates whether FG k overlaps with the interference-free window.
- $z_k^{\text{sense}} \in \{0,1\}$: Indicates whether FG k overlaps with the sensing window.
- $b_{\text{ws}} \in \{0,1\}$: Binary variable determining the relative ordering between the sensing window and the interference-free window.
- $b_{kj} \in \{0,1\}$: Ordering variable indicating whether FG k is scheduled before FG j.
- $y_k \geq 0$: Scheduling delay of FG k, consistent with Table 2.

(1) Linearized Objective and Dynamic Cost: The original objective is reformulated as a linear combination of decision variables:

$$\min \sum_{k=1}^{K} z_k^{\text{free}} \cdot d_k + \gamma \sum_{k=1}^{K} z_k^{\text{sense}} \cdot d_k + \lambda \sum_{k=1}^{K} y_k + \sum_{k=1}^{K} \mu_k \cdot P_k. \tag{11}$$

The scheduling delay is defined as:

$$\begin{aligned} y_k &= s'_k - s_k, \quad \forall k; \\ y_k &\geq 0, \quad \forall k. \end{aligned} \tag{12}$$

(2) Flexible Window Coordination and Mutual Exclusivity: The placement of the sensing and interference-free windows is governed by:

$$\begin{aligned} T_{\text{free}}^{end} &= T_{\text{free}}^{start} + D_{\text{free}}; \\ T_{\text{sensing}}^{end} &= T_{\text{sensing}}^{start} + D_{\text{sensing}}; \\ T_{\text{start}} &\leq T_{\text{free}}^{start},\ T_{\text{sensing}}^{start}; \\ T_{\text{free}}^{end},\ T_{\text{sensing}}^{end} &\leq T_{\text{end}}; \\ T_{\text{free}}^{end} &\leq T_{\text{sensing}}^{start} + M(1 - b_{\text{ws}}); \\ T_{\text{sensing}}^{end} &\leq T_{\text{free}}^{start} + M b_{\text{ws}}; \\ b_{\text{ws}} &\in \{0, 1\}. \end{aligned} \tag{13}$$

(3) Unified QoS and Postponement Logic: The scheduling feasibility under QoS constraints is expressed as:

$$\begin{aligned} a_k &\leq s'_k + M \cdot P_k, && \forall k; \\ s'_k + d_k &\leq e_k + M \cdot P_k, && \forall k; \\ T_{\text{end}} + d_k &\leq e_k + M \cdot (1 - P_k), && \forall k; \\ s'_k + d_k &\leq T_{\text{end}} + M \cdot P_k, && \forall k; \\ P_k &\in \{0, 1\}, && \forall k. \end{aligned} \tag{14}$$

(4) Inter-FG Conflict Avoidance: To prevent overlap between scheduled FGs:

$$\begin{aligned} s'_k + d_k &\leq s'_j + M(1 - b_{kj} + P_k + P_j), && \forall k \neq j; \\ s'_j + d_j &\leq s'_k + M(b_{kj} + P_k + P_j), && \forall k \neq j; \\ b_{kj} &\in \{0, 1\}, && \forall k \neq j. \end{aligned} \tag{15}$$

(5) Interference Detection and Sensing Protection: The overlap conditions with ZigBee-related windows are modeled as:

$$\begin{aligned} s'_k - T_{\text{free}}^{end} &\leq M \cdot z_k^{\text{free}}, \quad \forall k; \\ T_{\text{free}}^{start} - (s'_k + d_k) &\leq M \cdot z_k^{\text{free}}, \quad \forall k; \\ z_k^{\text{free}} &\in \{0, 1\}, \quad \forall k. \end{aligned} \tag{16}$$

$$\begin{aligned} s'_k - T_{\text{sensing}}^{end} &\leq M \cdot z_k^{\text{sense}}, \quad \forall k; \\ T_{\text{sensing}}^{start} - (s'_k + d_k) &\leq M \cdot z_k^{\text{sense}}, \quad \forall k; \\ z_k^{\text{sense}} &\in \{0, 1\}, \quad \forall k; \\ \sum_{k=1}^{K} z_k^{\text{sense}} &= 0. \end{aligned} \tag{17}$$

The final constraint enforces strict protection of the sensing window, ensuring that no WiFi transmission overlaps with it. This guarantees reliable cross-technology demand reporting, which is essential for the correct operation of CoWiC-MAC.

Data availability

Data will be made available on request.

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Tao Cheng received a bachelor's degree in software engineering and a master's in computer science and technology from Northwestern Polytechnical University in 2017 and 2020, respectively. He is currently a PhD candidate in computer science and technology from Northwestern Polytechnical University, Xi'an, China. His research interests include wireless communication, mobile computing, and heterogeneous wireless sensor networks.

Shining Li received the BS and MS degrees in computer science from Northwestern Polytechnical University, Xi'an, China, in 1989 and 1992, respectively, and the PhD in computer science from Xi'an Jiaotong University, Xi'an, China, in 2005. He is currently a professor at the School of Computer Science at Northwestern Polytechnical University. His research interests include wireless communication, mobile computing, and wireless avionics intra-communications.

Yan Pan received his Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2020. He is currently a Faculty with the National Key Laboratory of Information Systems Engineering, National University of Defense Technology since December 2020. His research interests include big data, machine learning, robotics, and mobile computing.